Table 4.2: Physical methods used to control microbial growth

	Method	Mechanism of action	Comment	Preferred for sterilizing
Heat				
1	**Dry heat**			
	a. Direct flaming	Burning contaminants to ashes	Very effective method of sterilization	Inoculating loops
	b. Incineration	Burning to ashes	Very effective method of sterilization	Paper cups, contaminated dressings, animal carcasses, bags, and wipes
	c. Hot –air sterilization	Oxidation	Very effective method of sterilization, but requires temperature of 160°C for about 1 hour	Empty glassware, instruments, needles, and glass syringes
2	**Moist heat**			
	a. Boiling or flowing steam	Protein denaturation	Kills vegetative bacterial and fungal pathogens and almost all viruses within 10 min; less effective on endospores	Dishes, basins, pitchers, various equipment
	b. Autoclaving	Protein denaturation	Very effective method of sterilization; at about 15 lbs of pressure (121°C), all vegetative cells and their endospores are killed in about 15 min	Microbiological media, solutions, linens, utensils, dressings, equipment, and other items that can withstand temperature and presure
	c. Pasteurization	Protein denaturation	Heat treatment for milk (72°C for about 15 sec) that kills all pathogens and most nonpathogens	Milk, cream, and certain alcoholic beverages(beer and wine)
3	**Radiation**			
	a. Ionizing	Destruction of DNA	Not widespread in routine sterilization	Used for sterilizing pharmaceuticals and medical and dental supplies
	b. Nonionizing	Damage to DNA	Radiation not very penetrating (non penetrating)	Control of closed environment with UV

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